Xuran (Alex) Sheng

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EDUCATION

University of Virginia, Charlottesville, VA *Expected May 2027*
Ph.D. in Economics

- **Dissertation Committee:** Prof. Eric Young (Chair), Prof. Mike Gallmeyer, Prof. Leland Farmer, Prof. Basil Halperin
- **Concentrations:** Asset Pricing, Portfolio Theory, Real Estate Finance, Computational Economics, Stochastic Control

University of Wisconsin–Madison, Madison, WI *August 2015 – May 2019*
B.A. in Mathematics, Computer Science, and Economics with Math Emphasis (Honors)

RESEARCH FIELDS

- **Primary Fields:** Asset Pricing, Portfolio Choice & Household Finance, Real Estate Finance
- **Secondary Fields:** Computational Methods, Stochastic Control, Machine Learning and Agentic AI

WORKING PAPERS & WORK IN PROGRESS

1. **"Infrequent Portfolio Adjustment and the Excess Volatility Puzzle"**
   (Job Market Paper)
   - *Abstract:* A lock-up is nearly free when the liquid sector is well-capitalized and expensive when it isn't, and most of that cost comes from the price distortion the constraint creates rather than from the inability to trade. The welfare cost is greater when the liquid and illiquid agents are further apart in their risk averseness.
2. **"Price to Rent and Mortgage Leverage"** (with Adriel Barretto)
3. **"Cost of Relocation, Renovation, and Commercial Real Estate Lease Terms"**
4. **"A Universal PINN Framework for Simulating Heterogeneous Agents in Constrained Markets"**
5. **"Rogue Agent: The Economics of Compute in the Agentic Age"**

RESEARCH & INDUSTRY APPOINTMENTS

Ph.D. Research Fellow, Global Places and Strategy *August 2024 – December 2025*
Nike, Inc., Beaverton, OR

- Developed a dynamic commercial property pricing model incorporating future rent escalation to guide executive real estate acquisition decisions.
- Analyzed workspace utilization, meeting space costs, and the relationship between employee co-location and in-person collaboration using high-frequency sensor and Microsoft Outlook telemetry data.
- Forecasted commercial lease dynamics by applying clustering methods and Vector Autoregressions (VAR) to historical corporate lease portfolios.
- Spearheaded AI/ML workflow integration and research automation across the Global Places and Strategy research team.

Graduate Research Assistant, Department of Economics *September 2019 – January 2023*
University of Virginia, Charlottesville, VA

- Collaborated with economics faculty to develop a dynamic life-cycle savings and portfolio choice model incorporating diverse retirement vehicles and annuities.
- Programmed high-performance numerical solvers in Julia and MATLAB utilizing backward induction, finite grid search, multidimensional interpolation, and Monte Carlo simulation.

Graduate Research Assistant, Department of Economics *August 2020 – August 2022*
University of Virginia, Charlottesville, VA

- Evaluated the welfare and risk-mitigation benefits of variable annuities relative to direct equity/bond investments via Monte Carlo simulation.
- Priced variable annuities, fixed index annuities, and registered index-linked annuities (RILAs) under alternative stochastic asset pricing frameworks to quantify post-retirement tail-risk mitigation.

Graduate Research Assistant, Darden School of Business *January 2021 – October 2021*
University of Virginia, Charlottesville, VA

- Investigated the endogenous relationship between financial intermediary development, returns to scale in innovation, and macroeconomic growth with finance faculty.

Graduate Research Assistant, School of Law *August 2020 – January 2021*
University of Virginia, Charlottesville, VA

- Examined post-retirement portfolio allocation and realized return performance with law and economics faculty.
- Built automated Python web-scraping pipelines and fuzzy string-matching algorithms to link multi-vendor mutual fund data with retiree portfolio holdings.

Undergraduate Research Assistant, CROWE *February 2017 – May 2019*
UW–Madison

- Constructed county-level employment panels in Stata and Excel to evaluate cross-border state policy spillovers on regional labor markets.

WEDC Research Assistant, Department of Economics *May 2018 – August 2018*
UW–Madison

- Estimated short- and long-run metropolitan trade and employment impacts of a proposed $10 billion manufacturing facility using MSA-level import/export regression analysis; built interactive web visualizations for the Wisconsin Economic Development Corporation.

TEACHING EXPERIENCE

Graduate Teaching Assistant *August 2020 – Present*
University of Virginia, Charlottesville, VA
Led 2–3 weekly discussion sections (~20 students per section), designed and graded problem sets and exams, and held regular office hours across the Department of Economics and Department of Mathematics:

- **MATH 4140:** Mathematics of Derivative Securities
- **MATH 4110:** Introduction to Stochastic Processes
- **ECON 4720:** Econometric Methods
- **ECON 3030:** Money and Banking
- **ECON 3010:** Intermediate Microeconomics *(Head Teaching Assistant)*
- **ECON 3020:** Intermediate Macroeconomics

Calculus Instructor *September 2016 – April 2017*
Greater University Tutoring Service, UW–Madison, Madison, WI

ACADEMIC SERVICE, HONORS & LEADERSHIP

Graduate Student Fellow, Quantitative Collaborative, University of Virginia *Mar. 2023 – Present*
Student Leader, Graduate Christian Fellowship (GCF), University of Virginia *Aug. 2022 – Present*
Graduate Representative, Student Council, University of Virginia *Jan. 2020 – Dec. 2020*
Team Captain, Pool and Billiards Club, University of Virginia *May 2019 – Present*

- *Winner, Collegiate 9-Ball Tour (2024, 2025); led team to Virginian Cup victories and coordinated post-COVID intercollegiate league revival.*

Presentation Team Member, Federal Reserve Challenge, UW–Madison *Jan. 2016 – Dec. 2018*
Board of Directors, University Book Store, UW–Madison *Oct. 2016 – May 2018*
Rec Sports Board, ASM Shared Governance Committee, UW–Madison *October 2016 – May 2017*

ADDITIONAL TECHNICAL EXPERIENCE

Technical & Statistical Support, SSCC, UW–Madison *July 2018 – May 2019*

- Provided statistical software consulting and computing infrastructure support to 1,000+ faculty and graduate researchers.

Scoreboard Operations Assistant, Intercollegiate Athletics, UW–Madison *Nov. 2015 – May 2019*

- Operated live broadcast replay, camera, and graphics systems for 80,000+ attendance events.

TECHNICAL SKILLS & LANGUAGES

- **Programming:** Python, Julia, MATLAB, R, SQL, Stata, C, FORTRAN, JavaScript, Java
- **Numerical Optimization & Solvers:** JuMP, Gurobi, MOSEK, Ipopt, Clp, Intel oneMKL
- **Machine Learning, Cloud & HPC:** PyTorch, TensorFlow, Scikit-learn, HPC (UVA Rivanna), Slurm, Linux, Git, Databricks, Snowflake, BigQuery, Microsoft Azure
- **Document & Media Tools:** LaTeX, Mathematica, Jupyter, HTML/CSS, Microsoft Office, Adobe Creative Suite
- **Languages:** Mandarin (Native), Spanish (Basic)

REFERENCES

Prof. Eric Young *(Chair)*
Carter Glass Professor of Economics
Department of Economics, University of Virginia
Email: ey2d@virginia.edu | Phone: (434) 924-3811

Prof. Mike Gallmeyer
Consumer Bankers Association Eminent Professor of Commerce
Director, Center for Investors and Financial Markets
McIntire School of Commerce, University of Virginia
Email: mgallmeyer@virginia.edu | Phone: (434) 243-4043

Prof. Marc Santugini *(Teaching Reference)*
Associate Professor
Department of Economics, University of Virginia
Email: ms3ae@virginia.edu | Phone: (434) 924-6756

Dr. Pilar Plater *(Professional Reference)*
Director, Analytics, Insights, & Research
Nike, Inc.
Email: Pilar.Plater@nike.com